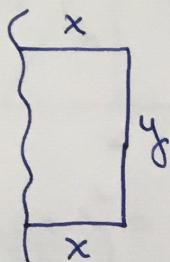


1.



$$2x + y = 400$$

$$y = 400 - 2x$$

Let x be the width and
 $400 - 2x$ be the length.

$$A = x(400 - 2x)$$

$$A' = 400 - 4x$$

$$A' = 0$$

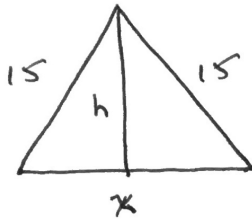
$$400 - 4x = 0$$

$$x = 100$$

$$\therefore \text{width} = 100 \text{ m}$$

$$\text{length} = 200 \text{ m.}$$

2.



$$h^2 + \left(\frac{x}{2}\right)^2 = 15^2$$

$$h^2 = 225 - \frac{x^2}{4}$$

$$h = \sqrt{225 - \frac{x^2}{4}}$$

$$A = \frac{1}{2} b \cdot h$$

$$A = \frac{1}{2} x \sqrt{225 - \frac{x^2}{4}}$$

$$A' = \frac{1}{2} \left[\sqrt{225 - \frac{x^2}{4}} - \frac{x^2}{4 \sqrt{225 - \frac{x^2}{4}}} \right]$$

$$A' = 0$$

$$4 \left(225 - \frac{x^2}{4} \right) - x^2 = 0$$

$$-2x^2 + 900 = 0$$

$$x^2 = 450$$

$$x \doteq 21.21 \text{ cm}$$

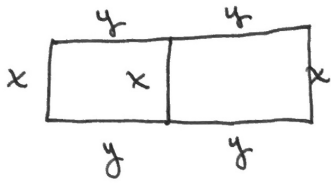
$$h \doteq 10.61 \text{ cm}$$

$$\therefore \text{Area} \doteq \frac{1}{2} (21.21)(10.61)$$

$$\text{Area} \doteq 112.5 \text{ cm}^2$$

(3)

3.



$$P = 3x + 4y$$

$$600 = 3x + 4y$$

$$y = \frac{600 - 3x}{4}$$

$$y = 150 - \frac{3x}{4}$$

$$A = L \cdot w$$

$$= x \left(150 - \frac{3x}{4} \right)$$

$$A' = 150 - \frac{3x}{2}$$

$$A' = 0$$

$$150 - \frac{3x}{2} = 0$$

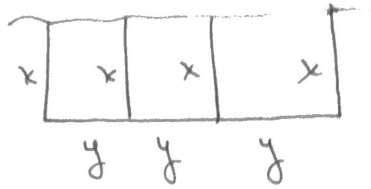
$$x = 100\text{m} \Rightarrow y = 75\text{m}.$$

$$\therefore \text{Area} = (100)(75)$$

$$\text{Area} = 7500 \text{ m}^2$$

(4)

4.



$$1600 = 4x + 3y$$

$$y = \frac{1600 - 4x}{3}$$

$$A = x \cdot y$$

$$A = x \left(\frac{1600 - 4x}{3} \right)$$

$$A' = \frac{1600}{3} - \frac{8x}{3}$$

$$A' = 0$$

$$\frac{1600}{3} - \frac{8x}{3} = 0$$

$$\therefore x = 200 \text{ m.}$$

$$y = \frac{1600 - 4(200)}{3}$$

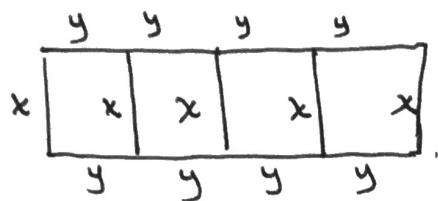
$$y = \frac{800}{3} \text{ for each field}$$

$$\text{Total} = 800 \text{ m.}$$

$$\therefore \text{Area} = (200)(800)$$

$$\text{Area} = 160,000 \text{ m}^2$$

5.



$$A = x \left(450 - \frac{5x}{8} \right)$$

$$a) P = 8y + 5x$$

$$3600 = 8y + 5x$$

$$y = \frac{3600 - 5x}{8}$$

$$A' = 450 - \frac{5x}{4}$$

$$A' = 0$$

$$450 - \frac{5x}{4} = 0$$

$$x = 360 \Rightarrow y = 225$$

each lot is 225m

Total length is
 $225 \times 4 = 900m$

$$A = (360)(900)$$

$$A = 324,000 m^2$$

b) must change y to 250m

$\Rightarrow x$ is now 320.

$$L = 250 \Rightarrow 8(250) = 2000$$

$$P = 3600 - 2000$$

$$W = 1600 \div 5$$

$$= 320m$$

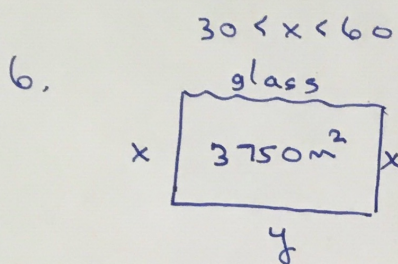
4 lots



$$\therefore A = (250 \times 320) \times 4$$

$$A = 320,000 m^2$$

(6)



$$A = xy$$

$$3750 = xy$$

$$y = \frac{3750}{x}$$

Let k be the cost per linear metre of cement wall.

$$\text{Total Cost} = kx + ky + kx + 2ky$$

$$\text{Total Cost} = 2kx + 3ky$$

$$\text{Total Cost} = 2kx + 3k \left(\frac{3750}{x} \right)$$

$$(\text{Total Cost})' = 2k - \frac{11250k}{x^2}$$

$$2k - \frac{11250k}{x^2} = 0$$

$$x^2 = 5625$$

$$x = 75 \Rightarrow y = 50$$

$\therefore$ Dimensions are

50 m glass &
75 m cement.

NOTE:

$$(\text{Total Cost})''$$

$$= 4kx + \frac{22500k}{x^3}$$

$$c(30) = 340k$$

$$c(50) = 300k$$

$$c(60) = 305k$$